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ABSTRACT

The Mars Rover has a lightweight integrated structure/insulation that has been environmentally tested and qualified for the 1996 launch for the Pathfinder mission to Mars. The basic structure with insulation, called the Warm Electronics Box (WEB), accounts for only 10% of the total Rover mass. The WEB is a thermal isolating composite structure with co-cured thermal control surfaces and an ultralightweight hydrophobic solid silica aerogel which minimizes conduction and radiation. This design provides excellent thermal insulation at low gas pressures and meets the structural requirements for spacecraft launch loads and for a 60 g impact landing at Mars without damage to the insulation or structure. The Rover design has been thermally qualified in a series of both static and transient thermal-vacuum testing in vacuum and 8 torr wind environments simulating all predicted environmental conditions. The Mars Rover has completed all environmental qualification and acceptance testing, and is undergoing system integration testing with the Pathfinder lander.

INTRODUCTION

The MicroRover Flight Experiment (MFEX) is part of the scientific and engineering payload for the Mars Pathfinder mission which will explore the Martian surface. The Pathfinder mission is intended to provide an engineering demonstration of key technologies and concepts for the future exploration of Mars employing surface landers and Rovers. The mission is scheduled for launch in December 1996 on a Delta II expendable launch vehicle and is expected to land on Mars on July 4, 1997. Upon arrival at Mars, the Pathfinder spacecraft will aerobrake through the Martian atmosphere protected by a heatshield, deploy a parachute, fire retrorockets, and inflate airbags to dissipate the remaining entry velocity for a landing on the Mars surface. The Mars Rover will then be deployed to explore the Mars surface. Its primary

exploration phase is to be completed in 7 Martian days, with an extended mission goal of 30 days. A picture of the full Rover system is shown in Figure 1. This paper will discuss the integrated thermal design of the Rover and the qualification testing in a simulated Martian surface environment.

The Mars Rover design was developed to meet varied thermal requirements, a low system mass allocation while maximizing functionality and a small stowed volume. The mobility system consists of six independently driven wheels and four steering mechanisms. It was limited to a total mobile mass of 11.0 kg, including the Alpha-Proton X-ray Spectrometer (APXS) scientific instrument, ten engineering experiment packages and an imaging system. The basic WEB structure is contained within a volume of 340 mm by 275 mm by 150 mm. The deployed Rover size is 650 mm by 470 mm by 320 mm. It has autonomous navigation, using a pair of wide angle imaging cameras for object recognition and hazard avoidance. Its mobility will be provided by a six wheel drive, rocker-bogie suspension system. It will maintain two-way contact with the Pathfinder lander via an ultra-high frequency (UHF) modem link with the lander. Power will be supplied from a 0.22 m² GaAs solar panel that will provide a peak power of 16 Watts and a non-rechargeable battery system that can provide 150 W-hr at 50 % depth of discharge. The Mars Rover thermal subsystem consists of a Warm Electronics Box (WEB), three RHU heater units to provide a constant 2.9 Watts heat input, and various heater elements on the batteries, modem and actuators. The WEB must vent the air pressure during launch, survive launch and landing, and meet strict planetary protection standards. The Rover also required approximately 150 wires to pass through the thermal enclosure to connect the solar array, motors, charge couple device (CCD) cameras, lasers, the drive system and other external experiments.

ENVIRONMENT

There are 5 distinct thermal environments which drove the design of the Mars Rover:

1. Earth Operations (no RHU's)
2. On the Launch pad (with RHU's)
3. Cruise (Radiation dominated vacuum)
4. First night on Mars while still attached to the cold finger (if necessary)
5. Mars exploration phase

Each environment posed different drivers on the Rovers thermal design. On Earth, convection and conduction dominated the thermal balance in a warm environment. During cruise, the thermal coupling to the relatively warm lander is nominally through radiation, since there is no convection in the vacuum environment and the launch attachments have a nonconductive link to the WEB. This drove the design for a conductive link from the Rover to the lander petal, which will have an active cooling system to remove waste heat during cruise^(1,2). After landing, the Rover may have to survive overnight attached in its stowed position to the Lander. In this scenario, the thermal link is still connected creating the coldest environmental conditions for the Rover. Additionally, Mars has an 8 torr CO₂ wind and a view to the cold sky.

The internal temperature range requirements are based on several constraints. The Mars Rover utilizes standard industrial and military electronic components which are qualified to operate between +/-55°C. The thermal design goal was limited to +/-40°C. The temperature limits are additionally constrained by the +/-30°C battery temperature requirements, with the allowable of up to 40°C for the 3 months at the launch pad and up to 55°C (with less than 5 hours per day above 40°C) on the Mars surface.

The proposed landing site for the Pathfinder mission is a rocky plain area known as Ares Vallis. The site is 850 kilometers (527 miles) southeast of the location of Viking Lander 1, at a latitude of 15 degrees North. For this northern latitude and arrival date of July 1997, the Lander will arrive late spring. There will nominally be a 12.5 hr period of sunlight, with a peak solar insolation ranging from 475 to 550 W/m², depending on the atmospheric opacity. The atmosphere is nominally 8 torr CO₂ gas, with the nominal average surface wind that may range from 0 to 3 m/s at night, and 2 to 8 m/s, with an average of 4 m/s during the day. Because the wind speed is influenced by drag force at the surface, the effective wind speed at the Rover level should be about half these levels⁽⁶⁾. The expected diurnal surface and atmospheric temperatures are expected to range from a high of -

13°C to a low of -89°C. The diurnal thermal profile is shown in Figure 2.

Mars provides a challenging thermal environment that is in a region of thermal transport that has not been well studied. It is also a regime which is not often encountered on terrestrial applications, and is very different from the vacuum environment all spacecraft encounter. The 8 torr CO₂ environment falls in a transition regime between the continuum regime (50 torr and above) and the Knudsen free molecular conduction regime (1 torr and below). There has been a limited amount of evaluation of insulation materials for these environments. Wilbert, *et.al.* conducted a study for NASA's 1976 Viking lander that evaluated foam insulation, fibrous, powders and multilayer insulation (MLI) for thermal control⁽³⁾. The result of that study was the primary use of several inches of foam insulation and some MLI on the Viking Landers. Other studies of thermal control in that time period assumed a foam insulation of 3 to 4 inches thick for potential Mars surface missions^(4,5). A more recent study revisited the issue of Mars thermal control and evaluated additional porous fiber and particulate insulation⁽⁶⁾. This bulky and mass intensive approach for thermal control continued for the Pathfinder lander, which uses 2 inches of 50 mg/cc foam enclosed in a composite honeycomb shell. All these systems or studies consider stationary landers and do not have the mass and volume limitations that the Rover does. Furthermore, larger landers have much more favorable thermal mass or volume to surface area ratio's, reducing difficulties with respect to cold. In addition, stationary landers have the benefit of greater power generation through deployed solar array panels. The MFEX Rover was the first system to fully consider the unique physical constraints of the Martian environment. Early in the program various insulation concepts such as foams, fibrous insulation, vacuum jacketed enclosures, and opacified powders were studied before settling on the use of 15 to 20 mg/cc monolithic silica aerogel for thermal control. The development and characterization of the aerogel design of the MFEX Warm Electronic Box (WEB) is discussed in Reference 7.

The use of aerogel for thermal control on Mars considers the thermal transport limitations for the transition between the continuum and free molecular regime. The effective thermal conductance of an insulative system consists of two components, the solid conductance-radiative component (independent of pressure) and the convective component. Gas conductance depends on the mobility inside the voids of a material, and is governed by the relative dimensions and connectivity of the open volume and the gas mean free path. If the interstitial space between material (whether they be voids, particles or fibers), becomes smaller than the mean free path of the

gas within the insulation, the mechanism for gas transport shifts from the continuum regime to free molecular conduction in the Knudsen regime. The mean free path of a gas is inversely proportional to the gas pressure. For CO₂ at 8 torr, the mean free path is approximately 3 to 5 microns in the temperature range from -90°C to 25°C. Solid aerogel has pore dimensions 6 to 11 nanometers. This means that the effective gas conduction within the aerogel is a fraction of the value in the continuum regime, and the thermal balance is reduced to the conductive-radiative component of the silica aerogel structure. Since low density aerogels are 80 to 90% open cells, direct conductance is minimized. Because of the high void fraction and the resulting weak mechanical properties of low density aerogels, thermal control had to be designed and integrated into the lightweight supporting structure. This basic integrated structure and thermal insulation design became the basis for the WEB. The thermal conductivity of the aerogel integrated structure design is listed in Table 1.

THERMAL DESIGN

The Mars Rover mission operation scenario is driven by the Martian environment and the power limitations on the Rover. Power is nominally provided by the solar array with batteries used for redundancy and reserve for emergency surge and nighttime operations. There are heaters on the batteries to enable heat storage inside the thermal enclosure. Since the motors can't start below about -75°C, heaters were installed for early morning operations (during operation the motors self heat as much as 40°C). There is also a film heater on the internal modem, to provide heating during early morning telecommunications. The navigation subsystem and other external experiments operate at the ambient surface conditions. No battery power will be used to maintain temperatures.

Primary internal heat is provided continuously by three 1-Watt radioisotope heating units (RHU's). To prevent the overheating of the Rover during cruise, a direct thermal link from the RHU's was incorporated into the thermal design. As illustrated in Figure 3, there is a direct conductive path from the structural axis tube to the Y-pin structural support. The path is via an aluminum clamp which is conductively bonded to the structural axle tube with silver-filled epoxy and mechanically attached to the aluminum accelerometer tray. Conductively coupled to the accelerometer tray is a copper braided strap which attached to a spring tensioned Y-pin. The tensioning spring provides 10 lb. of contact force to mating surface. Thermal contact resistance is reduced by having Choterm 11-79 between the Y-pin and the mating surface. This lander mounted surface is conductively coupled to the Lander active cooling loop⁽¹⁾ to dissipates approximately 2 watts continuously during the cruise stage of the

mission. When the Rover arrives at Mars, it stands up on its rocker-bogie mobility system (6 to 30 hours after landing), and breaks the thermal contact with the lander cold finger. A heater was installed on the Y-pin to compensate for the fluid loop should the Rover not deploy before the first night.

The silica aerogel was manufactured at the Jet Propulsion Laboratory (JPL) at a density of 15-20 mg/cc (0.94 -1.25 lb/ft³). Limited development time forced the design to have the WEB built around the Aerogel in a sequential manner, rather than integrally molded. Hence, the Rover utilized a sheet and spar design for the WEB. This integrated structural and thermal design is composed of a set of E-glass/epoxy structural members, with each volume filled with 2 blocks of low density solid silica aerogel insulation. The thickness and therefore strength of the facesheets and spars vary with location on the WEB to optimize stiffness and reduce mass. The facesheets vary from 12 to 20 mil glass/epoxy, and are co-cured with different films to provide the desired emissivity. The exterior surface is covered with gold coated Kapton to provide a low emissivity surface to reduce radiation, and the interior is covered with Ni coated graphite to provide a high emissivity surface to equalize temperatures. The Ni-coated graphite also provides EMI protection. Nominal aerogel thickness is 25 mm for the side walls, and 32 mm for the bottom wall. The solar array consists of a honeycomb sandwich panel with a 32 mm thick glass/epoxy covered solid aerogel plug which fits into the top of the WEB. The honeycomb cells directly above the WEB are also filled with aerogel. To reduce internal radiation, a 5 mil gold coated kapton film was placed between the two layers of solid silica aerogel to act as a radiation barrier. The basic structure and insulation, accounts for only 10% of the total Rover mass.

A structural design feature aiding the thermal design (by reducing spar thickness), is that the interior components provide structural support for the walls. The axle tube spans from wall to wall and provides the mechanical attachments for the rocker-bogie wheel assembly and transmits most of the WEB's launch and landing loads directly to the Lander. The RHU's are housed in the axle which also directly supports the batteries, boards and the accelerometer tray. It's made of E-glass/epoxy to meet structural requirements and minimize conductive losses. The axle is bonded to the walls with thermally isolating composite fittings. The boards also attach directly to the walls adding to the overall box rigidity.

Another thermal design feature of the WEB is the cable tunnel. The cable tunnel (seen at the front of the Rover in Figure 1) increases the thermal path length of the cabling that passes through the warm

enclosure. Loss down the cables is decreased by a factor of 15 by providing a 50 cm path as opposed to a direct 2.5 cm path. The insulation in the cable tunnel is fabricated from Eccofoam polyurethane foam.

QUALIFICATION TESTING

The Mars Rover design has been extensively tested, both for its mechanical and thermal performance. These tests were conducted at an engineering model level and at a flight System Integration Module (SIM) level and compared to the thermal model predictions. Reference 7 describes the engineering model testing and results, and provides a brief overview of the SINDA thermal model. The SIM qualification testing will be discussed here.

The objectives of the SIM Qual Thermal test was to demonstrate that the Rover maintains all components within operating and non-operating temperature ranges in a transient Martian environment. It was also intended to demonstrate functional performance at the Rover system level. The Rover SIM Qualification testing was conducted in a 10' vertical thermal vacuum test chamber at JPL. The test hardware was mounted inside with a wind generator system placed on the chamber end-bell (Figure 4). In addition to the chamber cold shroud there were three heat exchanger plates to simulate the ground (mounting plate), the solar/sky environment (sky plates) and a compensator plate to correct the gas temperature. The Rover SIM unit was situated on the mounting plate. When the Rover was deployed, the solar array was located about 12 inches beneath the sky plates, which consists of a copper (5'x3') and a stainless steel plate (5'x1'). During the initial phases the Rover, in the stowed mode, rested upon the Y-pin, which was connected to the fluid loop cold finger. To simulate Mars surface conditions, nitrogen at 8 torr was substituted for carbon dioxide. Nitrogen has a higher effective thermal conductivity, but testing limitations with the chamber (such as icing) prevented the use of CO₂. The SINDA thermal model was adjusted to account for the difference in the gasses.

The wind generating system consists of a 30" fan, a 5-hp motor, a motor housing with a ferrofluidic feedthrough, a motor cooling system, a flow straightener and supporting structure. The front end of the Rover faced the fan. Two reference plates (one gold and one black) were suspended above the Rover near the middle of the flow stream, to verify the environment and estimate wind speed.

The SIM QUAL testing had two distinct phases: A) SIM-Q1, a wind calibration stage followed by steady state thermal-vacuum testing which represented the cruise stage near Mars and; (B) SIM-Q2, a nominal transient environment phase (including the design

conditions for ground, air, and sky temperatures as well as wind) beginning with the Rover in the stowed position on the Lander petal until the second day, followed by 2 days of Mars surface operations. The test process was a simulation of the actual diurnal environment which starts after a 3:00 AM local time landing. The testing consisted of four independent but coordinated operations: (1) Rover functional operations which control the power dissipation at the Rover components (CPU/Power board, Modem, motor actuators, APXS electronics and current-limiter, and battery heaters), (2) chamber and environmental heat exchangers, (3) wind generator and (4) RHU & Y-pin heaters and reference-plate heaters. The coordination of these activities were based on a pre-determined time-event schedule.

During the test the Rover performed a number of pre-programmed tasks, including autonomous wakeup and shutdown, and Rover deployment. There were no anomalous temperature induced conditions. The effective wind near the reference plates was 1.0 m/s and was nominally 0.6 m/s in the zone adjacent to the Rover. The CPU/Power board dissipated around 3.77 watts during most of the Rover operating periods between wake-up and shut down. Figures 5 and 6 shows the temperatures for several of the interior components compared to the model results for the first day on Mars when the Rover is still conductively linked to the Lander. To help minimize the heat flow from the Rover, 2.73 Watts were applied to the Y-pin heater during the first day.

During Mars surface operations the minimum equipment temperatures are approximately 5°C warmer the corresponding levels with the Y-pin still connected. The interior components followed the board temperatures within a few degrees. The batteries, having the most thermal mass, lagged most in temperature (~7°C) as shown in Figure 7. The modem heater was cycled briefly and shown to have a short heat up time (spikes in Figure 7). The effect of the 0.6 m/s external wind (at the Rover height) increased the internal convection which lower the CPU temperature by about 8°C (Figure 8). Exterior components followed the simulated diurnal temperatures as expected (not shown).

SUMMARY

The design of the Mars Rover meets very constrained mass, volume and thermal requirements by utilizing low conductivity silica aerogel in an integrated structural design. The thermal environments requires the use of RHU's to survive the cold Martian nights and the integration of a thermal path to remove excess internal heat during cruise. In the nominal case the Rover will not have to provide additional heat to the batteries to survive the cold nights on Mars. The Mars

be launched in December 1996 for a July 1997 Mars landing.

ACKNOWLEDGMENTS

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Table 1. Thermal Conductivity measurements of the Aerogel with Glass-epoxy structure at the coupon level.

<u>Condition</u>	<u>Temperature</u> (°C)	<u>Thermal Conductivity</u> (W/m-K)
1 atm	25	0.0333
10 torr CO ₂	-79	0.0089
10 torr CO ₂	-27	0.0126
10 torr CO ₂	25	0.0163
1.2 x 10 ⁻³ torr Nitrogen	-78	0.0049
1.2 x 10 ⁻³ torr Nitrogen	-26	0.0074
0.4 x 10 ⁻³ torr Nitrogen	25	0.0113

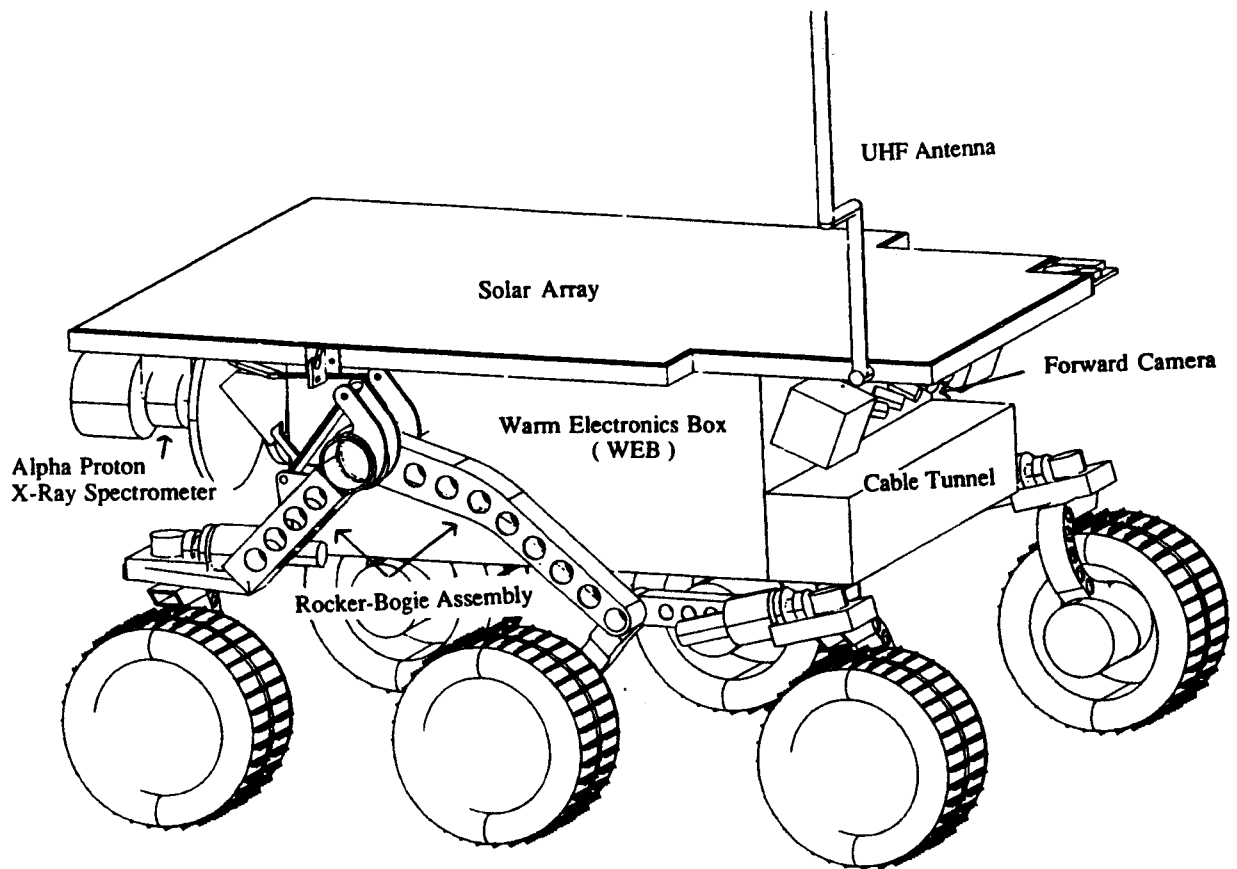


Figure 1. A schematic of the Mars Rover.

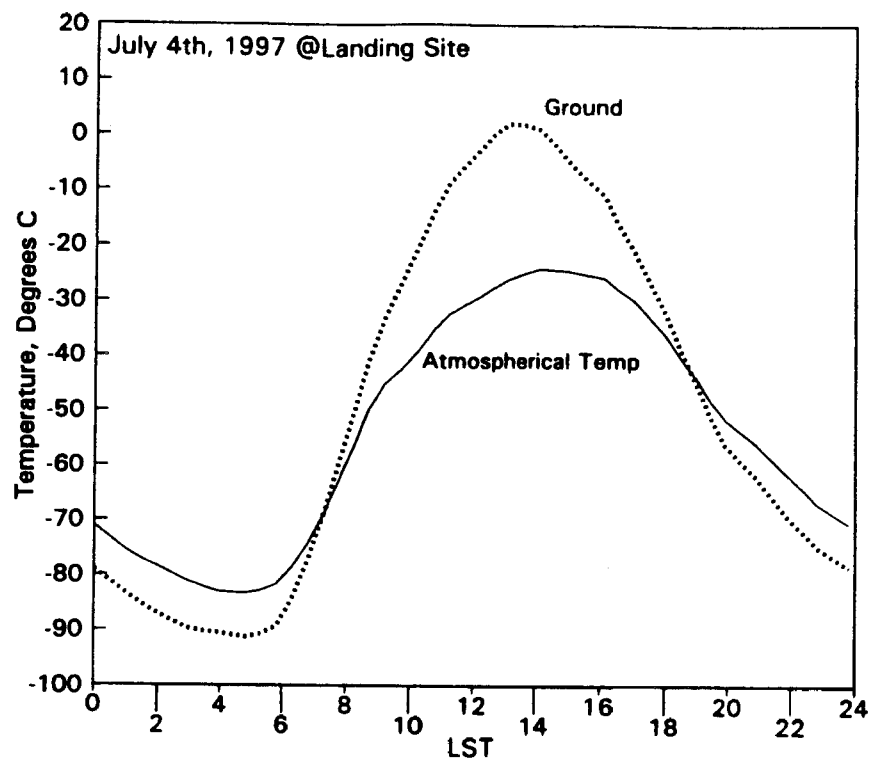


Figure 2. The diurnal thermal profile for the Mars Pathfinder landing site.

Figure 4. Test configuration for a the Mars Rover in a simulated Mars Environment including surface winds.

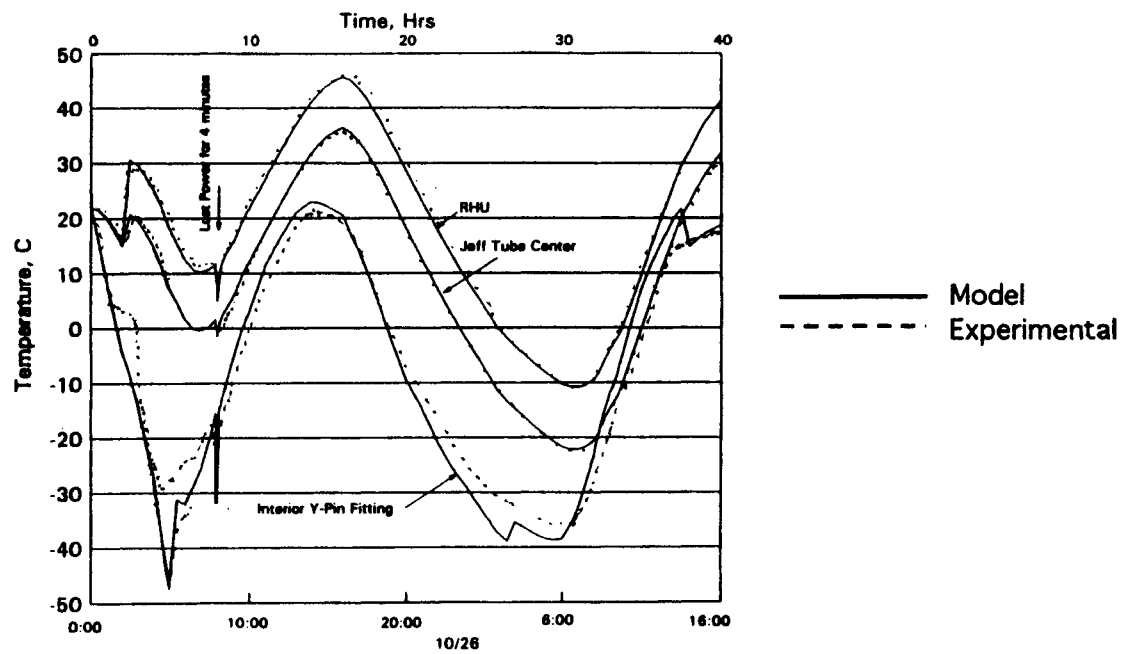


Figure 5. Predicted and experimental temperatures for internal straps during the first day on Mars when the Rover is still conductively attached to the Lander.

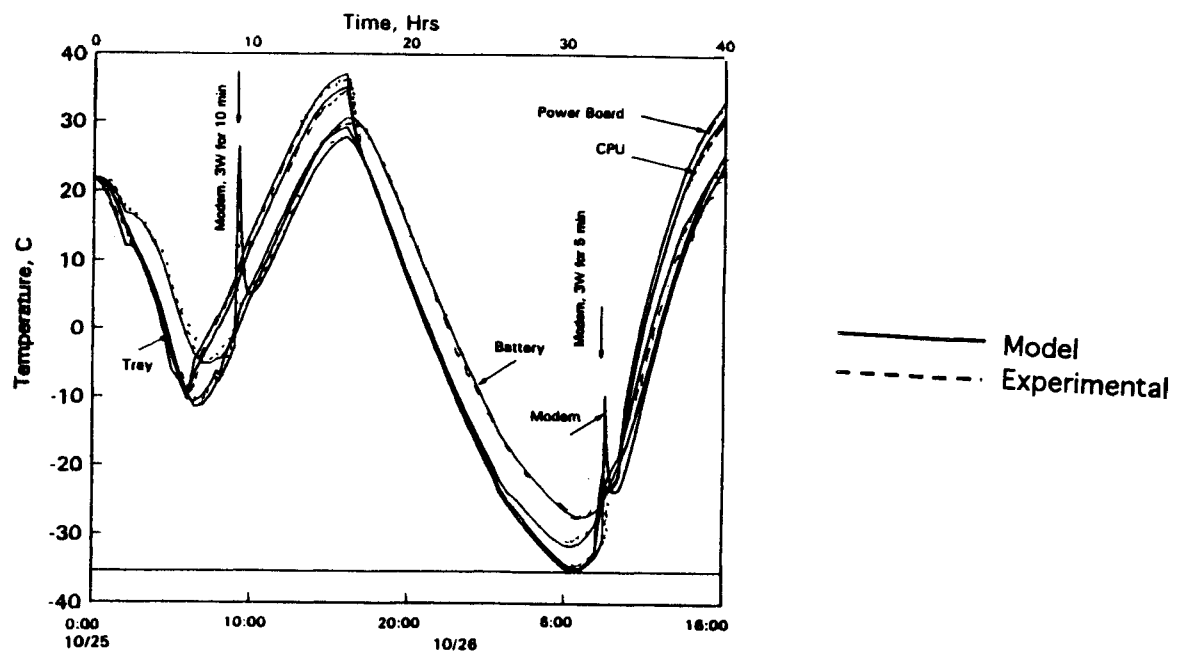


Figure 6. Predicted and experimental temperatures for interior components during the first day on Mars when the Rover is still conductively attached to the Lander.

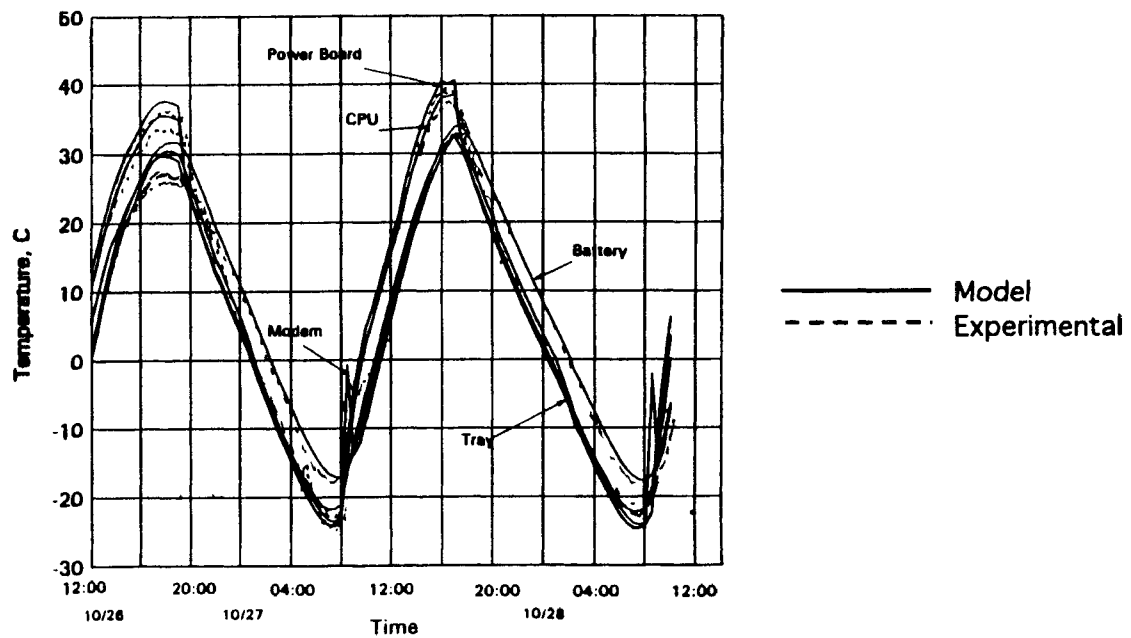


Figure 7. Predicted and experimental interior temperatures during the Mars surface operations with no wind.

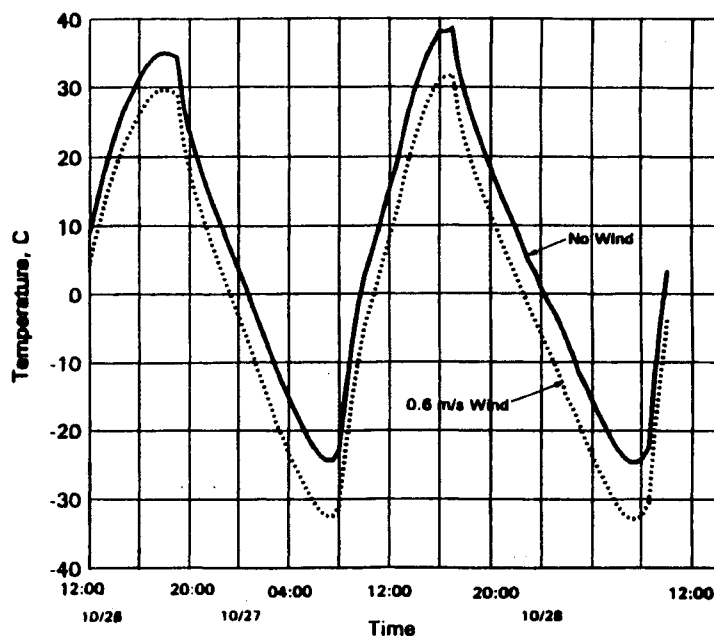


Figure 8. Effect of the Martian wind on the CPU board inside the warm electronics box.